Introduction to Photosynthesis

Photosynthesis is the process by which green plants use sunlight to make their own food. It takes place mainly in the leaves, inside tiny structures called chloroplasts. Chlorophyll is a green pigment that absorbs light energy. During photosynthesis, plants take in carbon dioxide from the air and water from the soil. They release oxygen as a by-product, which is essential for most living organisms on Earth. Scientists estimate that about 70 percent of the oxygen in our atmosphere comes from marine plants and algae.

The Light Reactions

The first stage of photosynthesis is known as the light-dependent reactions. These reactions occur in the thylakoid membranes of the chloroplast. Light energy is used to split water molecules into hydrogen and oxygen. The energy is stored temporarily in molecules called ATP and NADPH. Without sunlight, these reactions cannot happen, which is why plants grow slowly in dark places.

The Calvin Cycle

The second stage is called the Calvin cycle, and it does not need light directly. It uses the ATP and NADPH produced earlier to turn carbon dioxide into glucose. Glucose is a simple sugar that plants use for energy and growth. Extra glucose is stored as starch in roots, stems and seeds. Farmers can improve crop yields by making sure plants get enough light, water and carbon dioxide.